# HW 2 — Distributions and Maximum Likelihood

Mathematical Foundations of Diffusion Models · Monsoon 2026 · IIT Jammu

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| **Released** | Monday 10 August 2026 |
| **Due** | Tuesday 25 August 2026, in class |
| **Marks** | 50 points — graded on correctness |

**Graded on correctness.** Unlike HW 1, this set is marked on the work itself. Marks are carried by the *reasoning* as much as the final answer: a correct number with no derivation earns very little, and a wrong number reached by a sound argument earns most of the credit. Show every step you actually used. Solutions are posted after the deadline.

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**Notation used throughout.** Three things this sheet leans on.

- **i.i.d.** — *independent and identically distributed.* Saying $x_1, \ldots, x_n$ are i.i.d. from some distribution means each was drawn from the *same* distribution, and no draw influenced any other. It is what lets you multiply the individual probabilities together to get the probability of the whole sample.
- **Density vs likelihood** — the same formula, read in two directions. Fix the parameters and let the data vary and you have a *density* (or pmf): it describes how data comes out, and it integrates or sums to 1. Fix the observed data and let the *parameters* vary and you have the *likelihood*: a score saying how well each candidate parameter explains what you actually saw. It does *not* integrate to 1 over the parameters, and it is not a distribution over them. Maximum likelihood means: pick the parameter with the best score.
- $\Gamma$ **and** $B$ (Problem 1 only) — $\Gamma$ is the factorial extended to non-integers, and the only property you need is $\Gamma(z+1) = z\,\Gamma(z)$. The Beta function $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is just the constant that makes the Beta density integrate to 1.

Show your working. Where you set a derivative to zero, say why the critical point is a maximum and not a minimum.

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## Problem 1

A seed company sells lots of 500 seeds. Not every seed germinates, and the *germination rate* — the fraction of a lot that sprouts — is not the same number for every lot: soil, storage and harvest conditions all vary, so across lots the rate itself scatters somewhere in $[0, 1]$.

To model a quantity that is confined to $[0, 1]$ and varies, the standard choice is the **Beta** distribution. For shape parameters $\alpha, \beta > 0$ its density is

$$f(x; \alpha, \beta) \;=\; \frac{1}{B(\alpha, \beta)}\, x^{\alpha-1}\,(1-x)^{\beta-1}, \qquad 0 < x < 1,$$

and $f = 0$ outside $[0, 1]$, where

$$B(\alpha, \beta) \;=\; \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}\,dt \;=\; \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Larger $\alpha$ pushes the mass towards 1, larger $\beta$ towards 0, and both large together concentrates it. What the company wants to quote is the average germination rate and how much it varies from lot to lot — so, the mean and the variance.

Compute both *from the definition of expectation*: integrate against the density. Do not quote a formula.

(a) Show that $f$ integrates to 1. (One line, given the definition of $B$ above.) **(1pt)**

(b) Compute $\mathbb{E}[X] = \int_0^1 x\, f(x;\alpha,\beta)\, dx$. The trick throughout: multiplying by $x$ turns the integrand into another Beta kernel with $\alpha$ shifted, so the integral is a ratio of Beta functions. Then use $\Gamma(z+1) = z\Gamma(z)$. **(3pts)**

(c) Compute $\mathbb{E}[X^2]$ the same way. **(3pts)**

(d) Deduce $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ and simplify to a single fraction. **(2pts)**

(e) Check your mean at $\alpha = \beta$, and say why that value is obvious from the shape of the density without doing any integration. **(1pt)**

**Total: 10 points**

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## Problem 2

Probabilities are awkward to work with directly: they are trapped in $[0,1]$, so you cannot add to one without risking leaving the interval. The standard fix is to move to the **log-odds** scale. If an event has probability $u$, its odds are $u/(1-u)$ and its log-odds are

$$\mathrm{logit}(u) \;=\; \log\frac{u}{1-u},$$

which stretches $(0,1)$ out onto the whole real line. This is the *inverse* of the sigmoid $\sigma(z) = 1/(1+e^{-z})$, the function that has been squashing scores into probabilities all through this course.

Here is the question. Suppose a probability is itself uncertain, and completely unknown — model it as $U \sim \mathrm{Uniform}(0,1)$, every value equally plausible. Push it to the log-odds scale:

$$X \;=\; \mathrm{logit}(U).$$

Uniform noise on $(0,1)$ goes in; some distribution over all of $\mathbb{R}$ comes out. Which one?

(a) Verify that logit and $\sigma$ really are inverse: show $\sigma(\mathrm{logit}(u)) = u$. **(1pt)**

(b) Find the CDF of $X$ directly. Start from $F_X(x) = \mathbb{P}(X \le x)$, rearrange $\log\frac{U}{1-U} \le x$ into a statement about $U$ alone, and then use $\mathbb{P}(U \le c) = c$ for a uniform. **(3pts)**

(c) Differentiate to get the density $f_X$. Show it can be written in all three of these ways, and check it integrates to 1:

$$f_X(x) \;=\; \sigma(x)\big(1-\sigma(x)\big) \;=\; \frac{e^{-x}}{(1+e^{-x})^2} \;=\; \frac{1}{2\big(\cosh x + 1\big)}.$$

**(3pts)**

(d) Show $f_X(-x) = f_X(x)$, and deduce $\mathbb{E}[X]$ without integrating. **(2pts)**

(e) This distribution has a name — state it. Then explain in one or two sentences why the answer to (b) was a foregone conclusion before you did any algebra. **(1pt)**

*Part (e) is the real content. Feeding a uniform through the inverse of a CDF is precisely how one samples from that CDF — so applying $\sigma^{-1}$ to a uniform could only ever produce the law whose CDF is $\sigma$. This is the same "sampling is transport" idea the course uses when it writes a sample as $Q(U)$.*

**Total: 10 points**

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## Problem 3

A website is testing a new checkout page. It is shown to $n$ visitors, chosen independently of one another, and each either completes a purchase or does not. The conversion probability $p \in (0,1)$ is unknown — estimating it is the entire point of running the test.

Record $x_i = 1$ if visitor $i$ converts and 0 otherwise, so $x_1, \ldots, x_n$ are i.i.d. Bernoulli($p$), and let

$$S = \sum_{i=1}^{n} x_i$$

be the total number of conversions. Notice that the raw data is the whole sequence of 0s and 1s, but the number you would actually report is $S$.

(a) Write $\mathbb{P}(x_i = x)$ for $x \in \{0,1\}$ as a *single* expression in $p$ and $x$, with no cases. (This one line makes part (c) painless.) **(1pt)**

(b) Show that $S \sim \text{Binomial}(n,p)$, that is $\mathbb{P}(S = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Your argument needs three steps: why any one specific sequence with $k$ conversions has probability $p^k(1-p)^{n-k}$; how many such sequences there are; and why you are allowed to add their probabilities. **(3pts)**

(c) Write the likelihood $L(p)$ of the observed sequence and the log-likelihood $\ell(p)$. **(2pts)**

(d) Derive the maximum likelihood estimator $\hat{p}$ from $\ell'(p) = 0$, and confirm it is a maximum using $\ell''(p)$. Is the answer the one your intuition would have guessed? **(3pts)**

(e) Does anything change if you start instead from the Binomial likelihood for $S$, rather than the Bernoulli likelihood for the whole sequence? Say what happens to $\binom{n}{k}$, and why. **(1pt)**

**Total: 10 points**

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## Problem 4

A lab buys a second-hand instrument. Its datasheet is missing, so nobody knows the range over which it reports: readings are known to be spread evenly across some interval, but the endpoints of that interval are part of what has been lost.

Model $n$ readings as i.i.d. from Uniform($a, b$) with $a < b$ both unknown, and recover the calibration from the data.

(a) Write down the likelihood $L(a,b)$ for the whole sample. Take care: the density equals $1/(b-a)$ only *inside* the interval and 0 outside, so your expression must record that every observation lies in $[a,b]$. **(3pts)**

(b) Explain why setting $\partial\ell/\partial a = 0$ and $\partial\ell/\partial b = 0$ does *not* work here. **(2pts)**

(c) Argue directly from the shape of $L$ — not by differentiating — that

$$\hat{a} = \min_i x_i, \qquad \hat{b} = \max_i x_i.$$

**(3pts)**

(d) The estimated interval $[\hat{a}, \hat{b}]$ is always contained inside the true $[a,b]$. What does that say about whether $\hat{b} - \hat{a}$ is an unbiased estimate of the true width $b - a$, and in which direction it goes wrong? Would collecting more readings help? **(2pts)**

*Set this beside Problem 3. There, the maximum was found by calculus. Here calculus is the wrong instrument entirely and the answer falls out of the support constraint. Recognising which situation you are in is the skill being tested.*

**Total: 10 points**

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## Problem 5

A postal sorting machine reads the handwritten digit on an envelope and must decide which of the $K = 10$ classes $\{0, 1, \ldots, 9\}$ it is. From the image $x_i$ a network produces not one number but a *vector* of $K$ scores,

$$z_i \;=\; \big(z_{i1}, \ldots, z_{iK}\big) \;=\; f_\theta(x_i) \;\in\; \mathbb{R}^K,$$

one score per class, unconstrained in sign or size. These are called **logits**. To report probabilities they are passed through the **softmax**

$$q_{ik} \;=\; \frac{e^{z_{ik}}}{\sum_{m=1}^{K} e^{z_{im}}}, \qquad k = 1, \ldots, K.$$

The true label is written **one-hot**: $y_i = (y_{i1}, \ldots, y_{iK})$ with $y_{ik} = 1$ for the correct class and 0 for the other $K - 1$. So exactly one entry is a 1, and $\sum_k y_{ik} = 1$.

The machine is trained by minimising the **cross-entropy loss**

$$\mathrm{CE}(\theta) \;=\; -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} y_{ik}\, \log q_{ik}.$$

Like every loss in this course, it is not a formula to be accepted. It is a maximum-likelihood estimator wearing a different hat, and this problem takes the hat off.

(a) **Softmax is a distribution.** Show that $q_{ik} > 0$ for every $k$ and that $\sum_{k=1}^{K} q_{ik} = 1$, so the output really is a pmf over the $K$ classes. Then show that softmax is *shift-invariant*: adding the same constant $c$ to every logit leaves $q_i$ unchanged. What does that say about how many of the $K$ logits are genuinely free? **(2pts)**

(b) **The sampling distribution.** State the distribution of the label $y_i$ given the image $x_i$ that the machine is implicitly assuming, with $q_i$ as its parameter vector. Write its pmf as a *single* expression in $y_i$ and $q_i$, with no cases and no summation over classes — the one-hot encoding is what makes this possible. **(2pts)**

(c) **The generative model, and its assumptions.** Write the likelihood of the whole labelled dataset under that assumption. Take the logarithm, negate, divide by $n$, and show you land exactly on $\mathrm{CE}(\theta)$. Then state the three modelling assumptions you used: one about the *form* of each individual label's distribution, one about the relationship *across* the $n$ envelopes, and one about what role the images $x_i$ themselves play. **(3pts)**

(d) **The gradient.** For a single envelope, write $\ell(z) = -\sum_k y_k \log q_k$ with $q = \mathrm{softmax}(z)$. Show first that

$$\frac{\partial \log q_k}{\partial z_j} \;=\; \delta_{kj} - q_j, \qquad \delta_{kj} = \begin{cases} 1 & k = j \\ 0 & k \neq j \end{cases}$$

and deduce

$$\frac{\partial \ell}{\partial z_j} \;=\; q_j - y_j,$$

that is, predicted minus observed — in every coordinate at once. Say where in your derivation you used $\sum_k y_k = 1$. **(2pts)**

(e) **Back to two classes.** Set $K = 2$. Show that $q_2 = \sigma(z_2 - z_1)$, and hence that the loss above collapses to the binary cross-entropy of the lecture notes with a single logit $z = z_2 - z_1$. Which part of (a) explains why two logits were one too many? **(1pt)**

*Part (b) deserves the most thought. The machine is not generating envelopes — it is generating* labels*, conditionally on images it takes as given. That is why what you write down is a conditional sampling distribution, and why the likelihood that follows is a conditional one. Part (d) explains why this loss is so pleasant to optimise: the gradient is just the error, which is exactly the form the denoiser objective takes later in the course.*

**Total: 10 points**